Converting terminating decimals to fractions

Task A: Fractions less than 1

1) Convert the following into simplified fractions:

a) 0.65 $= \dfrac{13}{20}$

b) 0.34 $= \dfrac{17}{50}$

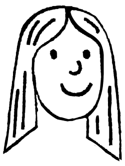
c) 0.12 $= \dfrac{3}{25}$

d) 0.542 $= \dfrac{271}{500}$

e) 0.864 $= \dfrac{108}{125}$

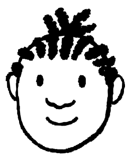
f) 0.1248 $= \dfrac{78}{625}$

2) Laura and Jacob are given 0.56 and 0.560 to convert into a fraction.



The answers will be different as 0.56 is two decimals places and 0.560 is 3 decimal places.

Laura says



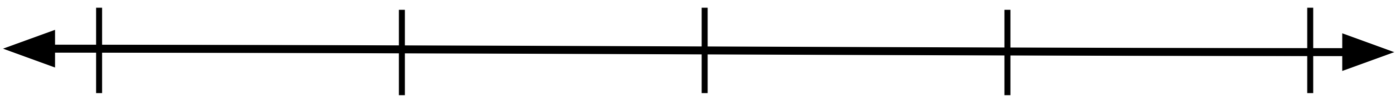
It is the same decimal so it will be the same fraction.

Jacob says

Who is correct and work out the fractional equivalent to the decimal(s)?

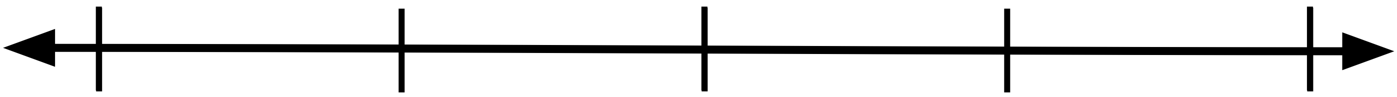
Both 0.56 and 0.560 are equal. They both are equivalent to $\dfrac{14}{25}$

3a) Write the fraction which is halfway between 0.96 and 0.88



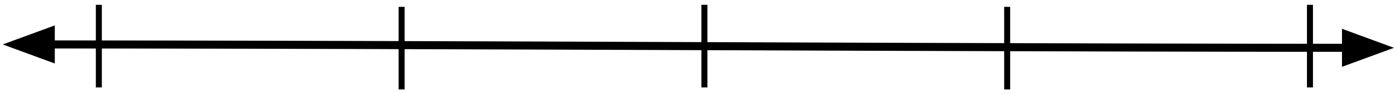
$\dfrac{23}{25}$

3b) Write the fraction which is a quarter of the way between 0.42 and 0.84

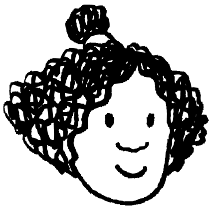


$\dfrac{21}{40}$

Three students created their own method to find the middle fraction.



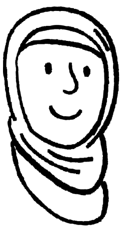
3c) Do all three methods work? Explain.



I converted 0.12 and 0.36 into fractions. You can see the middle easily.

Izzy

$0.36 = \frac{9}{25}$ *and* $0.12 = \frac{3}{25}$ *so the middle fraction is* $\frac{6}{25}$



Add 0.36 with 0.12, then halve the answer. Then find the fraction of the decimal.

Aisha

$0.36 + 0.12 = 0.48$ *then* $0.48 \div 2 = 0.24 = \frac{6}{25}$

Work out the difference between 0.36 and 0.12, then halve it and add it to 0.12. Convert this to a fraction.

Alex

$0.36 - 0.12 = 0.24$ *then* $0.24 \div 2 = 0.12$ *then* $0.12 + 0.12 = 0.24 = \frac{6}{25}$

Task B: Fractions greater than 1

1) Convert the following decimals into mixed numbers and improper fractions.

Decimal	Mixed number	Improper fraction
2.45	$2\frac{9}{20}$	$\frac{49}{20}$
8.125	$8\frac{1}{8}$	$\frac{65}{8}$
8.288	$8\frac{36}{125}$	$\frac{1036}{125}$
9.72	$9\frac{18}{25}$	$\frac{243}{25}$

2)

π is a special symbol and represents the non-terminating decimal 3.141592654…

There is no fractional equivalent to the exact value of π.

Work out the mixed number equivalent to:

a) 3.141 $= 3\frac{141}{1000}$

b) 3.1415 $= 3\frac{283}{2000}$

c) 3.141592 $= 3\frac{17699}{125\,000}$

Task C: Writing the denominator in exponential form

1) Work out the power of the exponent.

a) $0.4567 = \frac{4567}{10^4}$

b) $0.00017 = \frac{17}{10^5}$

c) $3.001 = \frac{3001}{10^3}$

2) Write the following decimals as a fraction where the denominator is in exponential form.

a) $0.1257 = \frac{1257}{10^4}$

b) $3.567 = \frac{3567}{10^3}$

c) $0.00023 = \frac{23}{10^5}$

3) Fill in the gaps.

a) $3.167 = \frac{3167}{1000} = \frac{3167}{10^3}$

b) $0.0357 = \frac{357}{10000} = \frac{357}{10^4}$

c) $0.15 = \frac{3}{20} = \frac{15}{100} = \frac{15}{10^2}$

d) $0.784 = \frac{98}{125} = \frac{784}{1000} = \frac{784}{10^3}$